

$$\begin{aligned}
& \max_{x \in \{0,1\}^n} \sum_{i=1}^n w_i x_i & (1) \\
& \text{s.t.} \quad \sum_{i=1}^n a_{ji} x_i \leq b_j & \forall j \in [m] \\
& & (2) \\
& x_i \in \{0,1\} & \forall i \in [n] \\
& & (3)
\end{aligned}$$